

Article

Design Approach on Bearing Capacity of the Cross-Bracing with Different Types of Joint Connection in Steel Lattice Transmission Towers

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Abstract: This paper presents an evaluation of the bearing capacity of cross-bracing in steel transmission tower structures. Design guidelines (ASCE 10-15, BS EN 50341-1, GB 50017-2017, and DL/T 5486-2020) related to the buckling capacity of the cross-bracing are summarized and compared with the experimental results. The current design provisions obtained the bearing capacity from the equivalent slenderness ratio, and then the stability coefficient and buckling capacity were derived. The calculated bearing capacity based on the design code tends to be overly progressive for smaller slenderness ratios (particularly those below 100), except for EN 50341-1-2012. Conversely, for larger slenderness ratios, ASCE 10-15 and DL/T 5486-2020 Class A design codes lean towards being overly progressive, while GB 50017-2017 and EN 50341-1-2012 codes tend to be more conservative. The design standard appears to exhibit unsafe predictions for Class A and B connections with low slenderness ratios and Class C connections. It needs to be noted that the effects of torsional stiffness and joint connection type are not considered in the current design codes, which are proved to be nonnegligible by the test results. In this paper, the bearing capacity calculation formula is proposed by introducing a modified effective length coefficient (K), and both the torsional stiffness and joint connection type are taken into account. The modified bearing capacity is verified with the test results; the correlation coefficient is 0.997, and the coefficient of variation is 0.04. It can provide a reference for the engineering design of steel lattice transmission tower structures.

Keywords: steel lattice transmission towers; cross bracing; slenderness ratio; torsional stiffness; design codes



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1. Introduction

Steel lattice transmission towers are vital lifeline projects to provide telecommunication or power transmission, as shown in Figure 1a. Consequently, their safety and reliability dominate the design [1–3]. In transmission tower structural engineering applications, angle steel components are predominantly interconnected using single-limb bolted connections, thereby constituting compression-bending elements [4,5]. Although this form of straightforward connection facilitates ease of construction and installation, the load-bearing conditions of the members are complex, primarily manifesting in two aspects. On one aspect, the load-bearing axis does not coincide with the principal centroidal axis of the cross-section, resulting in bidirectional eccentricity. This phenomenon, in turn, renders the determination of the bending axis direction during angle steel instability a nontrivial task [6]. On another aspect, the determination of end-constraint conditions poses challenges. As such, the single-angle steel compression strut, being the most frequently employed element within transmission towers, substantiates a pivotal control condition during angle

steel design, aiming to forestall instability under compressive loading. Axially loaded elements, when characterized by low slenderness ratios, are susceptible to flexural–torsional instability failure. Conversely, elements characterized by high slenderness ratios are prone to buckling-induced flexural instability failure. In transmission tower structures, diagonal members and leg members are typically connected with single-sided angle steel, with their load-carrying capacity being significantly influenced by eccentricity and constraint effects. Investigations into the ramifications of eccentricity have been undertaken extensively.

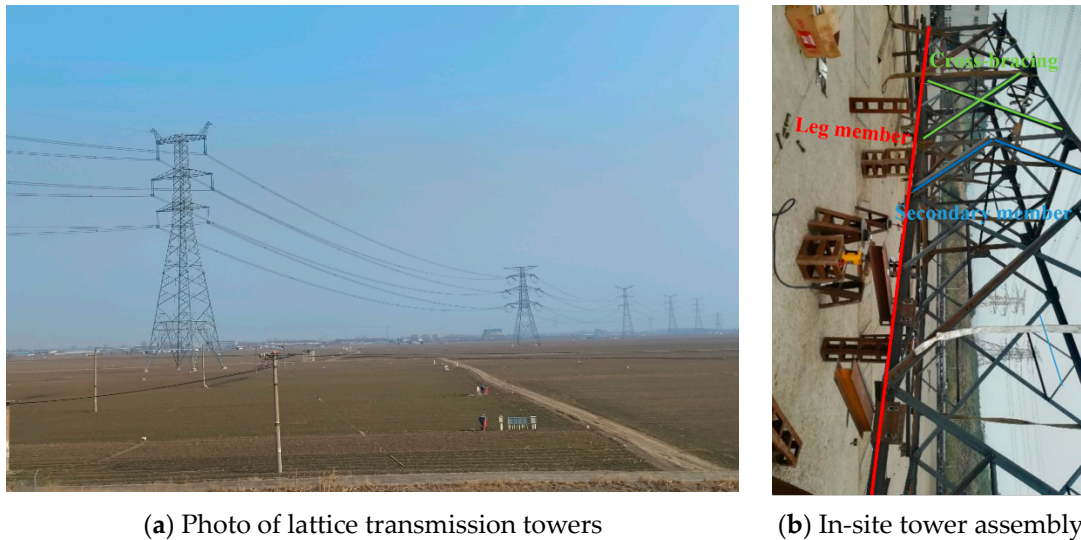


Figure 1. Schematic diagram of the members in lattice transmission towers.

Angle steel members tend to be in the eccentric compression state due to the single-side connection in engineering practice. With the increase in transmission capacity of the high-voltage grid, the joint connections have become more complex. Therefore, determining the stability capacity of the tower members is difficult [7]. The boundary conditions exhibited effects on the buckling behavior of the steel column. Fixed-ended and pin-ended equal-leg angle steel columns showed significantly different ultimate capacities [8,9]. They exhibited similar buckling performance: the critical load decreased with the length of the steel column, and a single half-wave mode was observed in terms of the critical buckling. For post-buckling behavior, both of them displayed the phenomenon that torsional rotations and corner displacement occurred simultaneously. Popovic et al. [10] performed cold-formed compression tests with boundary conditions fixed-ended and pin-ended, respectively. It was found that the experimental stub column strengths are 15% to 40% higher than the calculated results by the specifications. Moreover, Popovic et al. [11] conducted compression tests on cold-formed equal angles and concluded that the compression behavior is sensitive to the loading position; flexural–torsional buckling mode or flexural mode dominates the failure mode according to the loading position. Temple and Sakla [12,13] performed an investigation on the behavior of single-angle compression members welded by one leg to a gusset plate and concluded that the gusset plate thickness and the gusset plate width are the two most important parameters that affect the load-carrying capacity. Xu et al. [14] investigated the buckling behavior of aluminum alloy single-angle sections connected by one leg. Trahair [15] developed an approach to calculate the moment capacities of angle sections under combined bearing, shear, and torsion. The existing design codes (GB 50017-2017 [16], EN 1993-3-1 [17], and EN 50341-1 [18]) overestimated the buckling resistance of single-angle members with a small slenderness ratio, while the prediction by ASCE 10-15 [19] varies according to the different grades of the aluminum alloys. Figure 2 presents the schematic diagram of different typical buckling modes.

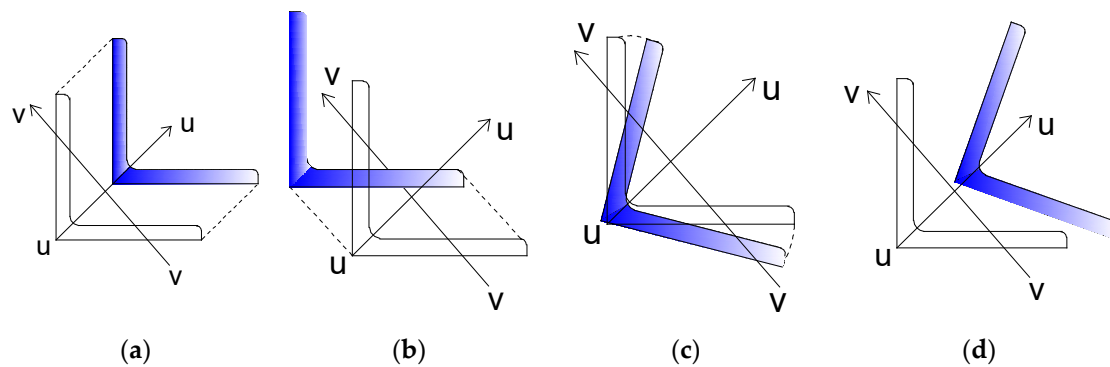


Figure 2. Schematic diagram of buckling modes [20]. (a) Flexural buckling along axis-uu; (b) flexural buckling along axis-vv; (c) torsional buckling; (d) flexural–torsional buckling.

The cross bracing is one of the primary load-bearing members of steel lattice transmission towers, constituting approximately 40% to 50% of the total tower weight, as shown in Figure 1b. Presently, there is a lack of established theoretical calculation formulas and a dearth of experimental data. Design standards for such members vary among different countries, with maximum calculated deviations reaching up to 43%. GB 50017-2017 [16] takes into consideration the effect of eccentricity while neglecting the favorable influence of constraints. ASCE 10-15 [19] provides modified formulas for the slenderness ratio, considering eccentricity and constraint effects separately. However, the boundary conditions only involve the number of bolts, with no explicit specification of nodal construction requirements. Europe identifies individual design guidelines and recommendations in various parts of Eurocode 3. More specifically, EN 1993-1-1 [21] defines the classification of four classes of cross-sections, and the resistance and rotation capacity are limited by their local buckling resistance. EN 50341-1-2012 [18] provides design rules for lattice transmission towers used in the field of overhead electrical lines. It introduces corrections through regularization of the slenderness ratio, which incorporates the constraint effect based on the number of bolts yet similarly lacks specific construction requirements. Due to the absence of experimental verification currently in engineering design, even for extra-high voltage towers such as 800 kV and 1000 kV, the number of bolts at the end of diagonal members can range from 4 to 6. The Chinese power industry standard (DL/T 5480-2020) [22] accounts for the constraint effects of two bolts. However, in many practical design scenarios, due to the consideration of differences in load-bearing methods for diagonal bracing and the lack of calculation theories, the influence of constraints has not been taken into account. The end restraints have a vital influence on the bearing capacity of angle steel members [23,24]. In current extra-high-voltage transmission tower projects, the end constraints for diagonal bracing generally involve five or more bolts and, in some cases, even up to ten bolts. The constraint effects of end bolts on members will be more pronounced. The substantial disparity in diagonal bracing calculation methods has led to conservative design results. Given the relatively high proportion of tower weight attributed to diagonal bracing, achieving optimization of diagonal bracing weight and reduction in tower weight through theoretical and experimental investigation is indeed a feasible endeavor. It should be noted that the literature focuses on the superstructure, and considering the foundation design, the corrosion of groundwater cannot be ignored [25,26].

Currently, in order to simplify the calculation of torsional buckling and flexural–torsional buckling of steel angle sections, standards mostly rely on determining the stability factors by calculating the equivalent slenderness ratio, thereby obtaining the stability bearing capacity of various types of steel angle sections. However, when equal-leg angle sections are subjected to axial compression, failure occurs due to bending along the weaker principal axis, while unequal-leg angle sections exhibit flexural–torsional buckling as their failure mode. In such cases, an equivalent slenderness ratio calculation approach

is employed. For the issue of local buckling, standards mostly consider local stability reduction factors, primarily related to the width-thickness ratio of the plate.

In this study, the recommended calculated methods of buckling capacity of a member in lattice transmission towers in different design codes are first detailed and summarized. Subsequently, a comparison between the experimental and theoretical results based on the design codes is discussed. The proposed correction factor of the slenderness ratio (K) considers the effects of torsional stiffness and joint construction type, which could guide the design of cross-bracing in steel transmission tower structures.

2. Existing Design Provisions

2.1. ASCE 10-15 Design of Latticed Steel Transmission Structures

The design compressive stress F_a on the gross cross-sectional area of axially loaded compression members is presented by Equations (1) and (2). The yielding stress of the steel f_y in Equation (1) is replaced by f_{cr} according to the value of $\frac{w}{t}$, where w = flat width and t = thickness of leg, as shown in Equation (3).

$$F_a = \begin{cases} \left[1 - \frac{1}{2} \left(\frac{KL/r}{C_c}\right)^2\right] f_y & KL/r \leq C_c \\ \frac{\pi^2 E}{(KL/r)^2} & KL/r > C_c \end{cases} \quad (1)$$

$$C_c = \pi \sqrt{2E/f_y} \quad (2)$$

$$f_{cr} = \begin{cases} \left[1.677 - 0.677 \frac{w/t}{80\psi/\sqrt{f_y}}\right] f_y & \frac{80\psi}{\sqrt{f_y}} \leq \frac{w}{t} \leq \frac{144\psi}{\sqrt{f_y}} \\ \frac{0.0332\pi^2 E}{(w/t)^2} & \frac{w}{t} > \frac{144\psi}{\sqrt{f_y}} \end{cases} \quad (3)$$

where $\psi = 1$ and 2.62 for f_y in ksi and MPa, respectively. E = modulus of elasticity; L = unbraced length; r = radius of gyration; and K = effective length coefficient.

2.2. BS EN 50341-1:2012 Overhead Electrical Lines Exceeding AC 1 kV-Part 1: General Requirements—Common Specifications

The compression angle should be verified against buckling, as in Equation (4). The design value of the compression force (N_{Ed}) should not exceed the design buckling resistance of the compression member ($N_{b,Rd}$). $N_{b,Rd}$ is presented in Equation (5), and χ is the non-dimensional buckling reduction factor. A is the gross area of the cross-section for Class 1, 2, and 3 sections, and A is the effective area of the cross-section for Class 4 section [21].

$$N_{Ed}/N_{b,Rd} \leq 1 \quad (4)$$

$$N_{b,Rd} = \chi A f_y / \gamma_{M1} \quad (5)$$

$$\chi = \frac{1}{\Phi + \sqrt{\Phi^2 - \bar{\lambda}_{eff}^2}} \quad (6)$$

$$\Phi = 0.5 \times [1 + \alpha \times (\bar{\lambda}_{eff} - 0.2) + \bar{\lambda}_{eff}^2] \quad (7)$$

where $\bar{\lambda}_{eff}$ is the effective non-dimensional slenderness, α is the imperfection factor and should be taken as equal to 0.13, and γ_{M1} is the partial resistance factor for the member.

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} \quad (8)$$

$$\bar{\lambda} = \begin{cases} \frac{\lambda}{\lambda_1} & \text{for Class 3 cross-section} \\ \frac{\lambda}{\lambda_1} \times \sqrt{\frac{A_{eff}}{A}} & \text{for Class 4 cross-section} \end{cases} \quad (9)$$

$\bar{\lambda}_{eff}$ is determined by $\bar{\lambda}$ and the types of members (leg members or bracing members).

2.3. GB 50017-2017 Chinese National Standard for Design of Steel Structures

According to GB 50017-2017, the stability of axial compression members satisfies Equation (10).

$$\frac{N}{\varphi A f_y} \leq 1.0 \quad (10)$$

where φ is the stability coefficient of the axial compression members, which can be referred to in the appendix to the design code. The stability coefficient is determined by the slenderness ratio, yield strength of the steel, and shape of the cross-section. A is the gross cross-sectional area of the member, and f_y is the designed tensile strength of the steel.

For singly connected single-angle steel columns, the stability of the specimen should be calculated by Equation (11). η is the reduction factor that considers the connection method and slenderness ratio, which can be calculated with Equation (12) for an equal-angle steel column.

$$\frac{N}{\eta \varphi A f_y} \leq 1.0 \quad (11)$$

$$\eta = 0.6 + 0.0015\lambda \quad (12)$$

2.4. DL/T 5486-2020 Chinese Power Industry Standard

The bearing capacity of angle steel satisfies Equation (13), where φ is the stability coefficient of axial compression members, m_N is the strength reduction factor of the compressive column and can be calculated with Equation (14), ω is the outstanding width of flange, t is the thickness of the flange, and η_c is the maximum ratio of flat width to the thickness and can be calculated with Equation (15). The effective length coefficient K is introduced to modify the slenderness ratio, which is the same as the definition in Table 1 in ASCE 10-15.

$$\frac{N}{m_N \varphi A f_y} \leq 1.0 \quad (13)$$

$$m_N = \begin{cases} 1.0 & (\frac{\omega}{t} \leq \eta_c) \\ 1.677 - 0.677 \frac{\omega}{t \eta_c} & (\frac{\omega}{t} > \eta_c) \end{cases} \quad (14)$$

$$\eta_c = 13 \sqrt{\frac{235}{\varphi f_y}} \quad (15)$$

Table 1. Correction factor of slenderness ratio for compression members.

Loading Conditions	Slenderness Ratio	Effective Length Coefficient K
Axial compression at both ends	$0 < L/r < 120$	1
Axial compression at one end and eccentric compression at the other end	$0 < L/r < 120$	$0.75 + 30/(L/r)$
Eccentric compression at both ends	$0 < L/r < 120$	$0.5 + 60/(L/r)$
No boundary at either end	$120 \leq L/r \leq 200$	1
Boundary at one end	$120 \leq L/r \leq 225$	$0.762 + 28.6/(L/r)$
Boundary at both ends	$120 \leq L/r \leq 250$	$0.615 + 46.2/(L/r)$

3. Comparison between the Experimental and Calculated Results

Experimental results in the literature [27] are used to make comparisons between the test and calculated results from design standards. Three types of connections are used to simulate different boundary conditions, as shown in Figure 3. For Class A connection, the test specimen is directly connected to the leg member with two bolts, resulting in good restraint effects. For Class B connections, the test specimen is connected to the leg member via a gusset plate. One bolt is directly connected to the leg member, resulting in a larger

eccentricity but allowing for a reduction in gusset plate dimensions. Class C connections also employed a gusset plate to link the test specimen to the leg member, with the smallest eccentricity but with a larger gusset plate. The test parameters of three types of specimens are presented in Table 2. The Class A connection is suitable for larger leg member joints, while the Class B and Class C connections are suitable for joints with smaller leg member specifications. The experimental results are shown in Table 3.

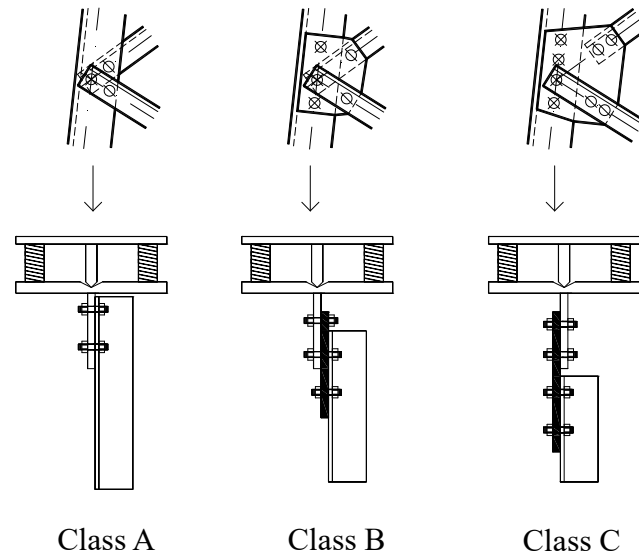


Figure 3. Schematic of end restraints.

Table 2. Test parameters for specimens. Data from [27].

Type	Angle Size	Thickness of Gusset Plate	Material of Column	Material of Plate	Yield Strength (MPa)
Class A	L80 × 7	-	Q355	Q355	373.23
Class B	L80 × 7	10	Q355	Q355	368.55
Class C	L80 × 7	10	Q355	Q355	367.64

3.1. Buckling Bearing Capacity

The geometrical properties, geometrical axis, and principal axis are shown in Figure 4, and the axis is defined as follows:

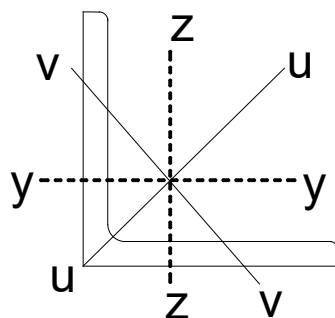


Figure 4. Notations of the steel angle section. yy/zz: geometric axis of the cross-section parallel to the leg; uu: major/strong principal axis (associated with weak axis displacement); vv: minor/weak principal axis (associated with strong axis displacement).

Table 3. The comparison between experimental and calculated results.

Notation	Torsional Stiffness (kN·m/rad)	Buckling Axis	Slenderness Ratio	Bearing Capacity (kN)						Comparison				
				Test Results	ASCE 10-15	EN 50341-2012	GB 50017-2017	DL/T 5486-2020 Class a	DL/T 5486-2020 Class b	ASCE/Test	EN/Test	GB/Test	DL-a/Test	DL-b/Test
AK2-60	50	vv	60	163.2	251.2	193.8	224.9	214.5	188.7	1.54	1.19	1.38	1.31	1.16
AK2-80	50	vv	80	147.7	215.1	153.5	181.7	181.8	161.9	1.46	1.04	1.23	1.23	1.10
AK2-100	50	vv	100	134.4	178.4	123.7	136.3	154.7	139.4	1.33	0.92	1.01	1.15	1.04
AK2-130	50	vv	130	104.9	135.7	92.4	91.3	121.3	110.9	1.29	0.88	0.87	1.16	1.06
AK2-160	50	vv	160	76.6	103.3	71.5	65.5	94.3	87.4	1.35	0.93	0.85	1.23	1.14
AK1-80	0	yy/zz	80	106.3	213.9	152.2	180.9	180.7	160.8	2.01	1.43	1.70	1.70	1.51
AK1-100	0	yy/zz	100	89.4	177.6	122.7	136.0	153.9	138.6	1.99	1.37	1.52	1.72	1.55
AK1-120	0	yy/zz	120	66.9	149.2	100.7	103.4	132.0	120.1	2.23	1.50	1.54	1.97	1.79
AK1-150	0	yy/zz	150	58.4	112.1	77.0	72.6	101.7	93.9	1.92	1.32	1.24	1.74	1.61
AK1-180	0	yy/zz	180	44.8	87.3	60.7	54.1	80.5	75.1	1.95	1.35	1.21	1.79	1.67
AK2-80	50	yy/zz	80	148.0	213.9	152.2	180.9	180.7	160.8	1.45	1.03	1.22	1.22	1.09
AK2-100	50	yy/zz	100	122.3	177.6	122.7	136.0	153.9	138.6	1.45	1.00	1.11	1.26	1.13
AK2-120	50	yy/zz	120	114.5	149.2	100.7	103.4	132.0	120.1	1.30	0.88	0.90	1.15	1.05
AK2-150	50	yy/zz	150	78.0	112.1	77.0	72.6	101.7	93.9	1.44	0.99	0.93	1.30	1.20
AK2-180	50	yy/zz	180	68.9	87.3	60.7	54.1	80.5	75.1	1.27	0.88	0.79	1.17	1.09
AK3-80	100	yy/zz	80	147.7	213.9	152.2	180.9	180.7	160.8	1.45	1.03	1.22	1.22	1.09
AK3-100	100	yy/zz	100	130.6	177.6	122.7	136.0	153.9	138.6	1.36	0.94	1.04	1.18	1.06
AK3-120	100	yy/zz	120	113.4	149.2	100.7	103.4	132.0	120.1	1.32	0.89	0.91	1.16	1.06
AK3-150	100	yy/zz	150	96.7	112.1	83.1	72.6	101.7	93.9	1.16	0.86	0.75	1.05	0.97
AK3-180	100	yy/zz	180	79.3	87.3	65.4	54.1	80.5	75.1	1.10	0.82	0.68	1.01	0.95
BK2-80	50	yy/zz	80	129.7	209.8	150.5	180.7	177.5	158.2	1.62	1.16	1.39	1.37	1.22
BK2-100	50	yy/zz	100	109.8	173.9	121.2	136.0	151.0	136.1	1.58	1.10	1.24	1.37	1.24
BK2-120	50	yy/zz	120	85.6	146.1	99.3	103.4	129.5	117.9	1.71	1.16	1.21	1.51	1.38
BK2-150	50	yy/zz	150	63.4	109.8	76.0	72.5	99.7	92.1	1.73	1.20	1.14	1.57	1.45
BK2-180	50	yy/zz	180	48.9	85.5	59.8	54.1	78.8	73.6	1.75	1.22	1.11	1.61	1.51
BK3-80	100	yy/zz	80	152.2	209.8	150.5	180.7	177.5	158.2	1.38	0.94	1.19	1.17	1.04
BK3-100	100	yy/zz	100	120.2	173.9	121.2	136.0	151.0	136.1	1.45	0.96	1.13	1.26	1.13
BK3-120	100	yy/zz	120	94.6	146.1	99.3	103.4	129.5	117.9	1.54	1.00	1.09	1.37	1.25
BK3-150	100	yy/zz	150	74.1	109.8	76.0	72.5	99.7	92.1	1.48	0.97	0.98	1.34	1.24
BK3-180	100	yy/zz	180	54.8	85.5	59.8	54.1	78.8	73.6	1.56	1.04	0.99	1.44	1.34
CK2-80	50	yy/zz	80	205.8	209.8	150.5	180.7	177.5	158.2	1.02	0.73	0.88	0.86	0.77
CK2-100	50	yy/zz	100	148.3	173.9	121.2	136.0	151.0	136.1	1.17	0.82	0.92	1.02	0.92
CK2-120	50	yy/zz	120	114.7	146.1	99.3	103.4	129.5	117.9	1.27	0.87	0.90	1.13	1.03
CK2-150	50	yy/zz	150	70.2	109.8	76.0	72.5	99.7	92.1	1.56	1.08	1.03	1.42	1.31
CK2-180	50	yy/zz	180	50.0	85.5	59.8	54.1	78.8	73.6	1.71	1.20	1.08	1.58	1.47
CK3-80	100	yy/zz	80	180.3	209.8	150.5	180.7	177.5	158.2	1.16	0.83	1.00	0.98	0.88
CK3-100	100	yy/zz	100	135.3	173.9	121.2	136.0	151.0	136.1	1.29	0.90	1.01	1.12	1.01
CK3-120	100	yy/zz	120	100.4	146.1	99.3	103.4	129.5	117.9	1.45	0.99	1.03	1.29	1.17
CK3-150	100	yy/zz	150	81.0	109.8	76.0	72.5	99.7	92.1	1.35	0.94	0.90	1.23	1.14
CK3-180	100	yy/zz	180	50.6	85.5	59.8	54.1	78.8	73.6	1.69	1.18	1.07	1.56	1.45

Figure 5a presents a comprehensive comparison between experimental and design codes for members buckling along the minor axis-vv. It is observed that both GB 50017-2017 and DL/T 5486 Class b consistently yield values lower than the corresponding experimental results. Specifically, GB 50017-2017 demonstrates a range of 4% to 44% lower than the test results. This significant deviation from the test values can be attributed to the lack of consideration for the contribution of end constraints in members with large slenderness ratios. Furthermore, GB 50017-2017 exhibits a higher level of conservatism regarding the impact of eccentricity on bearing capacity reduction compared to other design codes. On the other hand, calculated results by DL/T 5486 Class b are consistently 6% to 18% lower than the experimental values. In terms of DL/T 5486 Class a and EN 50341-1-2012, it is observed that for slenderness ratios below 120, their respective calculated results exceed the experimental results by approximately 4% to 10%. However, for slenderness ratios exceeding 120, the calculated results are lower than the test values by 7% to 23%. It is worth noting that the deviation between DL/T 5486 Class a and experimental results is relatively smaller compared to EN 50341-1-2012. For ASCE 10-15, it is found that when the slenderness ratio is below 160, it surpasses the test values by 2% to 21%. However, for slenderness ratios greater than 160, ASCE 10-15 falls below the test values by approximately 6%. For smaller slenderness ratios, DL/T 5486-2020 Class a, EN 50341-1-2012, and ASCE 10-15 tend to be overly progressive, with test results consistently lower than the calculated results. As the slenderness ratio increases, the test results gradually exceed the calculated results. During the design process, the DL/T 5486-2020 Class b is recommended, as it is relatively conservative and may require certain corrections. GB 50017-2017 yields overly conservative results because only eccentricity is considered and the effect of constraints is neglected.

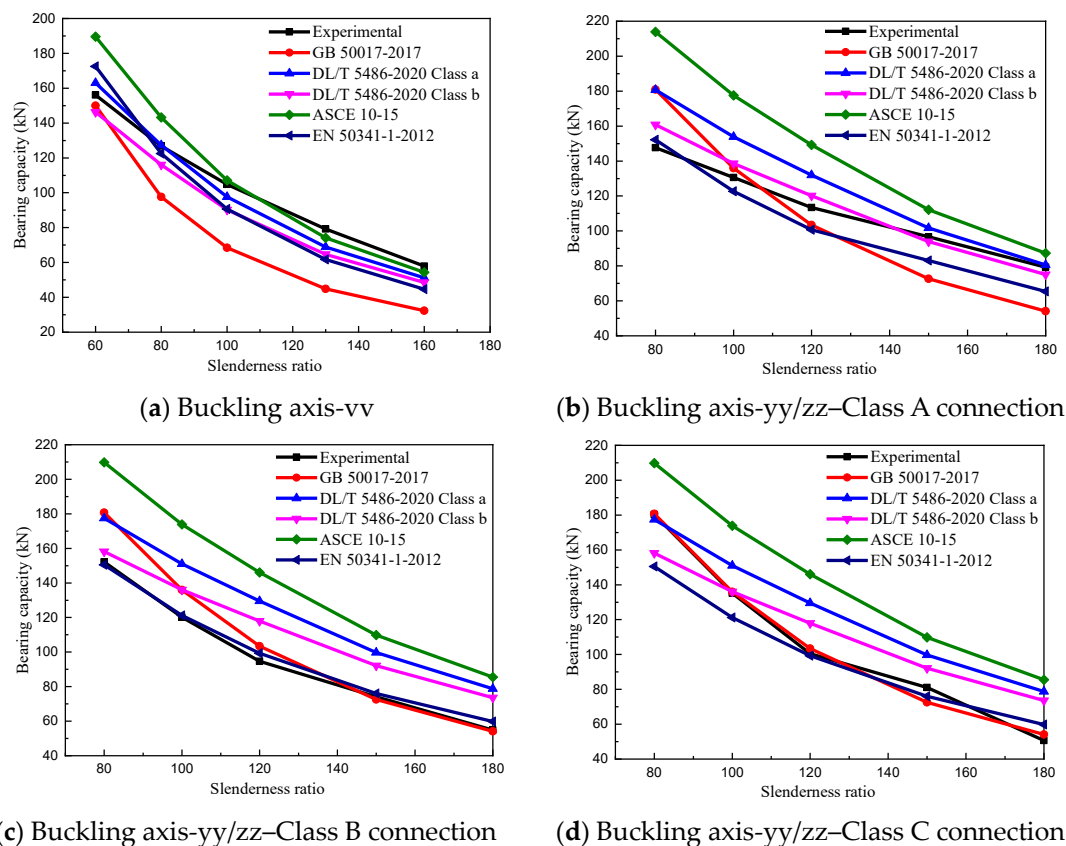


Figure 5. Comparison between experimental and calculated results with different end restraints.

Figure 5b presents the bearing capacity curve for the buckling axis-yy/zz with a Class A connection. Calculated results by DL/T 5486-2020 Class a and ASCE 10-15 consistently

surpass the experimental results, with DL/T 5486-2020 Class a exceeding the test results by 1% to 22% and ASCE 10-15 surpassing them by 10% to 45%. For a slenderness ratio below 100, GB 50017-2017 yields values higher than the experimental results, ranging from 4% to 22%, whereas for slenderness ratios higher than 100, it falls below the test results by 9% to 32%. When the slenderness ratio surpasses 80, EN 50341-1-2012 produces values lower than the experimental results, ranging from 6% to 18%. Regarding the DL/T 5486-2020 Class b, when the slenderness ratio is below 120, it exhibits values higher than the experimental results by 6% to 9%, while for slenderness ratios exceeding 120, it falls below the test results by 3% to 5%. Except for EN 50341-1-2012, the calculated results tend to be overly progressive for smaller slenderness ratios, particularly those below 100. Conversely, for larger slenderness ratios, the ASCE 10-15 and DL/T 5486-2020 Class a design codes lean towards being overly progressive, while the GB 50017-2017 and EN 50341-1-2012 codes tend to be more conservative. Therefore, during the design process, adopting the DL/T 5486-2020 Class b guidelines is recommended. It tends to be overly progressive for smaller slenderness ratios but can be appropriately adjusted, aligning closely with experimental results for larger slenderness ratios.

Figure 5c illustrates the bearing capacity curve for the buckling axis-yy/zz with a Class B connection. Apart from the GB 50017-2017 design code, which falls below the test results by less than 4% for slenderness ratios higher than 120, all other results by design codes are higher than experimental results. When DL/T 5486-2020 Class b code is used in the design process, it is observed that calculated results are higher than experimental results by 4% to 34%, and certain corrections are necessary, particularly for members with larger slenderness ratios.

Figure 5d presents the bearing capacity curve for the buckling axis-yy/zz with a Class C connection. Calculated results by the DL/T 5486-2020 Class a and ASCE 10-15 codes surpass the test results, with DL/T 5486-2020 Class a exceeding them by 12% to 56% and ASCE 10-15 exceeding them by 16% to 69%. EN 50341-1-2012 and GB 50017-2017 exhibit closer agreement with the test results, with an approximate 10% deviation. When employing DL/T 5486-2020 Class b during the design process, it is found that calculated results are 10% lower than the test results for slenderness ratios below 100, while for slenderness ratios exceeding 100, calculated results exceed the test results by 14% to 45%, indicating the need for appropriate adjustments.

3.2. Column Curve

Figure 6 presents a comparison between experimental results and column curves in the design standard. Column curves “a,” “b,” “c,” and “d” represent various types of cross-sections [16]. The buckling axis of member AK2-vv is weak principal axis-vv, and λ_n denotes the normalized slenderness ratio. It can be observed that when λ_n is less than or equal to 1.5, the experimental results closely align with the column curve “c”. As the slenderness ratio increases, there is a gradual transition from curve “c” to curve “a.” When λ_n exceeds 1.75, it surpasses curve “a.” In the case of the buckling axis-yy/zz for the Class A connection, experimental results transition from the curve “d” to the curve “a” when the slenderness ratio is in the range of 1.0 to 2.0. When λ_n is greater than or equal to 2.0, the experimental data surpass the curve “a.” The Class B connection primarily approximates the curve “c” and gradually approaches the curve “d” as the slenderness ratio increases. The stability coefficient for members with Class C connections moves from curve “b” to curves “c” and “d” as the slenderness ratio increases. It can be seen that angle steel sections with different types of joint connections and slenderness ratios show different variation trends. Notably, the design standard appears to exhibit unsafe predictions for Class A and B connections with low slenderness ratios and Class C connections. As a consequence, a refined formula is proposed to modify the slenderness ratio.

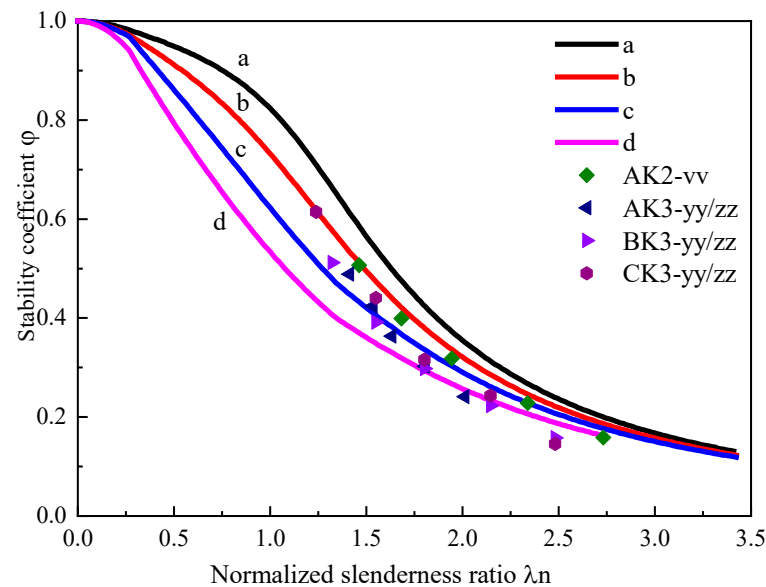


Figure 6. The comparison between test results and the column curve in the design code.

4. Recommended Design Guidelines

Currently, when it comes to torsional buckling and flexural–torsional buckling of the angle steel column, the standards calculate the stability bearing capacity by determining the equivalent slenderness ratio and corresponding stability factors. For local buckling issues, standards usually incorporate the local stability reduction factor, which is primarily related to the width-to-thickness ratio of the angle steel plate. The experimental and calculated results are compared by the design standards, and fitting of the slenderness ratio is conducted on three different types of joint connections and buckling modes by the least squares method. The recommended calculation formulas for the effective length coefficient (K) are presented in Equations (16)–(19). The joint connection, buckling modes, and torsional stiffness are taken into account in the formula. The definition of L/r is the same as that in Equation (1). K_0 is the torsional stiffness of the joint. It can be concluded from the previous study [27] that eccentricity is the dominant factor that affects the bearing capacity when L/r is less than 120, while end restraints have a more significant effect on the capacity when L/r is greater than 120. Therefore, a piecewise formula is proposed, considering the value of L/r .

- Buckling axis: weak principal axis-vv

$$K = \begin{cases} 0.5 + \frac{60}{L/r} \cdot \left(\frac{K_0}{L/r}\right)^{0.1} & 0 < L/r \leq 120 \\ 0.615 + \frac{46.2}{L/r} \cdot \left(\frac{K_0}{L/r}\right)^{0.16} & L/r > 120 \end{cases} \quad (16)$$

- Buckling axis: geometric axis of the cross-section parallel to the leg-yy/zz

Class A Connection

$$K = \begin{cases} 0.5 + \frac{60}{L/r} \cdot \left(\frac{K_0}{L/r}\right)^{0.1} & 0 < L/r \leq 120 \\ 0.615 + \frac{46.2}{L/r} \cdot \left(\frac{K_0}{L/r}\right)^{0.16} & L/r > 120 \end{cases} \quad (17)$$

Class B Connection

$$K = \begin{cases} 0.5 + \frac{60}{L/r} \cdot \left(\frac{L/r}{K_0}\right)^{0.1} \cdot 1.07 & 0 < L/r \leq 120 \\ 0.615 + \frac{46.2}{L/r} \cdot \left(\frac{L/r}{K_0}\right)^{0.1} & L/r > 120 \end{cases} \quad (18)$$

Class C Connection

$$K = \begin{cases} 0.5 + \frac{60}{L/r} \cdot \left(\frac{L/r}{K_0}\right)^{0.1} \cdot 1.05 & 0 < L/r \leq 120 \\ 0.615 + \frac{46.2}{L/r} \cdot \left(\frac{K_0}{L/r}\right)^{0.1} & L/r > 120 \end{cases} \quad (19)$$

GB 50017-2017 considers the reduction factor η when calculating the stability-bearing capacity. DL/T 5486-2020 takes a different approach from GB 50017-2017 and adjusts the slenderness ratio to account for eccentricity and confinement effects. The comparison between the experimental and calculated bearing capacity by the proposed effective length coefficient (K) is shown in Table 4 and Figure 7. It can be seen that the modified calculated results match the experimental results well. The correlation coefficient is 0.997, and the coefficient of variation is 0.04.

Table 4. Bearing capacity of test results and calculated results.

Slenderness Ratio	Bearing Capacity (kN)								
	Class A Connection			Class B Connection			Class C Connection		
	Test	DL/T 5486-2020 Class b	DL/T 5486-2020 Class b-Modified	Test	DL/T 5486-2020 Class b	DL/T 5486-2020 Class b-Modified	Test	DL/T 5486-2020 Class b	DL/T 5486-2020 Class b-Modified
80	147.7	160.8	144.2	152.2	158.2	142.5	180.3	158.2	172.5
100	130.6	138.6	120.1	120.2	136.1	110.1	135.3	136.1	120.1
120	113.4	120.1	104.9	94.6	117.9	88.4	100.4	117.9	88.4
150	96.7	93.9	88.4	74.1	92.1	64.8	81.0	92.1	64.8
180	79.3	75.1	73.0	54.8	73.6	49.6	50.6	73.6	49.7

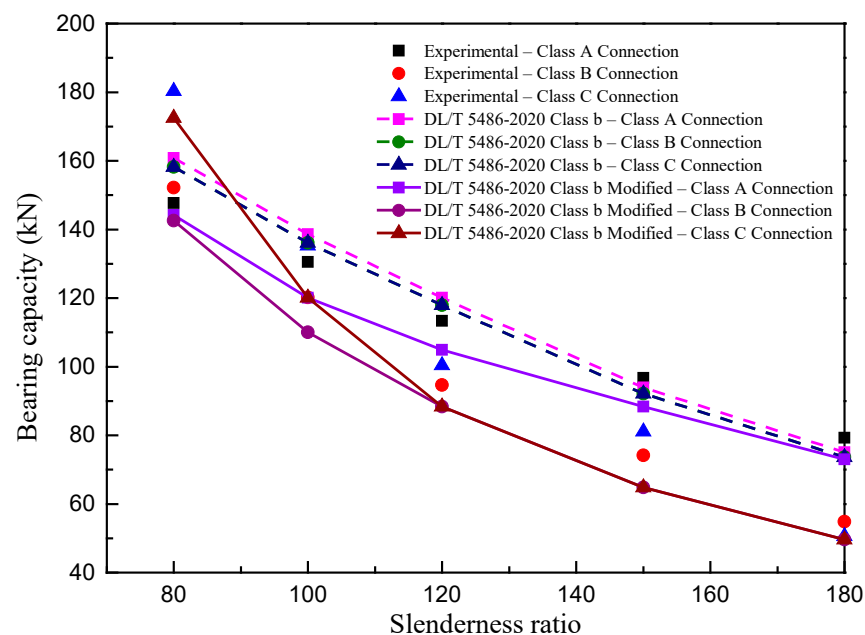


Figure 7. The comparison between experimental and modified results.

5. Conclusions

The current design codes simplify the cross bracing as an axial compression member for stability capacity calculation. The effects of eccentricity and rotational constraint are considered when determining the effective slenderness ratio. However, the influence of torsional stiffness and joint connection type is neglected in the current design codes. As a result, a comparative study reveals that the design code values exhibit discrepancies when compared to experimental results. Except for EN 50341-1-2012, the bearing capacity calculated by design codes tends to be overly progressive for smaller slenderness ratios, particularly those below 100. Conversely, for larger slenderness ratios, ASCE 10-15 and DL/T 5486-2020 Class a design codes lean towards being overly progressive, while GB

50017-2017 and EN 50341-1-2012 codes tend to be more conservative. Usually, in practical design, three types of joint connections exist. All four design standards appear to exhibit unsafe predictions for Class A and B connections with low slenderness ratios. The discrepancy is primarily manifested in eccentricity for small slenderness ratio columns and inadequate restraint corrections for large slenderness ratio columns, indicating certain limitations in the current approach. The proposed modified effective length coefficient (K) considers the effect of joint connection (Class A, B, and C) and torsional stiffness and fits well with the experimental results. The correlation coefficient is 0.997, and the coefficient of variation is 0.04.

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References

- de Souza, R.R.; Miguel, L.F.F.; McClure, G.; Alminhana, F.; Kaminski, J., Jr. Reliability Assessment of Existing Transmission Line Towers Considering Mechanical Model Uncertainties. *Eng. Struct.* **2021**, *237*, 112016. [\[CrossRef\]](#)
- Chatterjee, A.; Bosco, E.; Rajagopalan, S.; Varghese, I.M.; Ramalingam, R. A Review of Research on Supported Transmission Line Tower Failure Studies: Analysis, Tower Testing and Retrofitting. *Int. J. Steel Struct.* **2024**, *24*, 264–279. [\[CrossRef\]](#)
- Xie, Q.; Zhang, J. Experimental Study on Failure Modes and Retrofitting Method of Latticed Transmission Tower. *Eng. Struct.* **2021**, *226*, 111365. [\[CrossRef\]](#)
- An, L.; Wu, J.; Jiang, W. Experimental and Numerical Study of the Axial Stiffness of Bolted Joints in Steel Lattice Transmission Tower Legs. *Eng. Struct.* **2019**, *187*, 490–503. [\[CrossRef\]](#)
- Bezas, M.-Z.; Antonodimitraki, S.; Beyer, A.; Van Wittenberghe, J.; Anwaar, M.-O.; Maesschalck, J.; Jaspart, J.-P.; Demonceau, J.-F. Buckling Resistance of Star-Battened Angle Members Made of High-Strength Steel. *Structures* **2023**, *58*, 105274. [\[CrossRef\]](#)
- Xu, J.; He, S.; Cao, X.; Yao, L.; Liu, X.; Yan, X. Analytical and Numerical Analysis of Angle Steel Joints for Conductors in Lattice Transmission Towers. *Case Stud. Constr. Mater.* **2022**, *17*, e01592. [\[CrossRef\]](#)
- Li, Y.; Yan, Z.; Zhou, D.; Ren, Q.; Hua, C.; Zhong, Y. Analysis of the Stability Behavior of Cross Bracings in Transmission Towers Based on Experiments and Numerical Simulations. *Thin-Walled Struct.* **2023**, *185*, 110554. [\[CrossRef\]](#)
- Dinis, P.B.; Camotim, D. A Novel DSM-Based Approach for the Rational Design of Fixed-Ended and Pin-Ended Short-to-Intermediate Thin-Walled Angle Columns. *Thin-Walled Struct.* **2015**, *87*, 158–182. [\[CrossRef\]](#)
- Silvestre, N.; Dinis, P.B.; Camotim, D. Developments on the Design of Cold-Formed Steel Angles. *J. Struct. Eng.* **2013**, *139*, 680–694. [\[CrossRef\]](#)
- Popovic, D.; Hancock, G.J.; Rasmussen, K.J.R. Axial Compression Tests of Cold-Formed Angles. *J. Struct. Eng.* **1999**, *125*, 515–523. [\[CrossRef\]](#)
- Popovic, D.; Hancock, G.J.; Rasmussen, K.J.R. Compression Tests on Cold-Formed Angles Loaded Parallel with a Leg. *J. Struct. Eng.* **2001**, *127*, 600–607. [\[CrossRef\]](#)
- Temple, M.C.; Sakla, S.S. Single-Angle Compression Members Welded by One Leg to a Gusset Plate. I. Experimental Study. *Can. J. Civ. Eng.* **1998**, *25*, 569–584. [\[CrossRef\]](#)
- Temple, M.C.; Sakla, S.S. Single-Angle Compression Members Welded by One Leg to a Gusset Plate. II. A Parametric Study and Design Equation. *Can. J. Civ. Eng.* **1998**, *25*, 585–594. [\[CrossRef\]](#)
- Xu, Z.; Tong, G.; Zhang, L.; Guo, Y. Testing, Modelling and Design of Aluminium Alloy Single-Angle Members Connected by One Leg under Eccentric Compression. *Thin-Walled Struct.* **2023**, *184*, 110474. [\[CrossRef\]](#)
- Trahair, N.S. Bearing, Shear, and Torsion Capacities of Steel Angle Sections. *J. Struct. Eng.* **2002**, *128*, 1394–1398. [\[CrossRef\]](#)
- GB 50017-2017; Standard for Design of Steel Structures. China Architecture & Building Press: Beijing, China, 2018.

17. EN 1993-3-1; Eurocode 3—Design of Steel Structures—Part 3-1: Towers, Masts and Chimneys—Towers and Masts. European Committee for Standardisation: Brussels, Belgium, 2006.
18. BS EN 50341-1:2012; Overhead Electrical Lines Exceeding AC 1kV-Part 1: General Requirements—Common Specifications. European Committee for Standardisation: Brussels, Belgium, 2012.
19. ASCE 10-15; Design of Latticed Steel Transmission Structures. American Society of Civil Engineers: Reston, VA, USA, 2015.
20. Bezas, M.-Z. *Design of Lattice Towers from Hot-Rolled Equal Leg Steel Angles*; National Technical University of Athens: Athens, Greece, 2021.
21. BS EN 1993-1-1:2005; Eurocode 3-Design of Steel Structures:Part 1-1: General Rules and Rules for Buildings. European Committee for Standardisation: Brussels, Belgium, 2005.
22. DL/T 5486-2020; Technical Specification for the Design of Steel Supporting Structures of Overhead Transmission Line. China Planning Press: Beijing, China, 2020.
23. Jiang, W.Q.; Wang, Z.Q.; McClure, G.; Wang, G.L.; Geng, J.D. Accurate Modeling of Joint Effects in Lattice Transmission Towers. *Eng. Struct.* **2011**, *33*, 1817–1827. [[CrossRef](#)]
24. Jiang, W.-Q.; Liu, Y.-P.; Chan, S.-L.; Wang, Z.-Q. Direct Analysis of an Ultrahigh-Voltage Lattice Transmission Tower Considering Joint Effects. *J. Struct. Eng.* **2017**, *143*, 04017009. [[CrossRef](#)]
25. Xiao, J.; Huang, L.; He, Z.; Qu, W.; Li, L.; Jiang, H.; Zhong, Z.; Long, X. Probabilistic Models Applied to Concrete Corrosion Depth Prediction under Sulfuric Acid Environment. *Measurement* **2024**, *234*, 114807. [[CrossRef](#)]
26. Xiao, J.; Zeng, H.; Huang, H.; Liu, L.; Li, L.; Yuan, B.; Zhong, Z. Experimental Investigation on the Influence of Strength Grade on the Surface Fractal Dimension of Concrete under Sulfuric Acid Attack. *Buildings* **2024**, *14*, 713. [[CrossRef](#)]
27. Liu, H.; Xu, X.; Huang, Z.; Zhang, Y.; Huang, X.; Li, Z.; Guo, Y. Stability of Equal-Leg Angle Section Braces with Torsional Restraints. *J. Constr. Steel Res.* **2023**, *201*, 107717. [[CrossRef](#)]

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